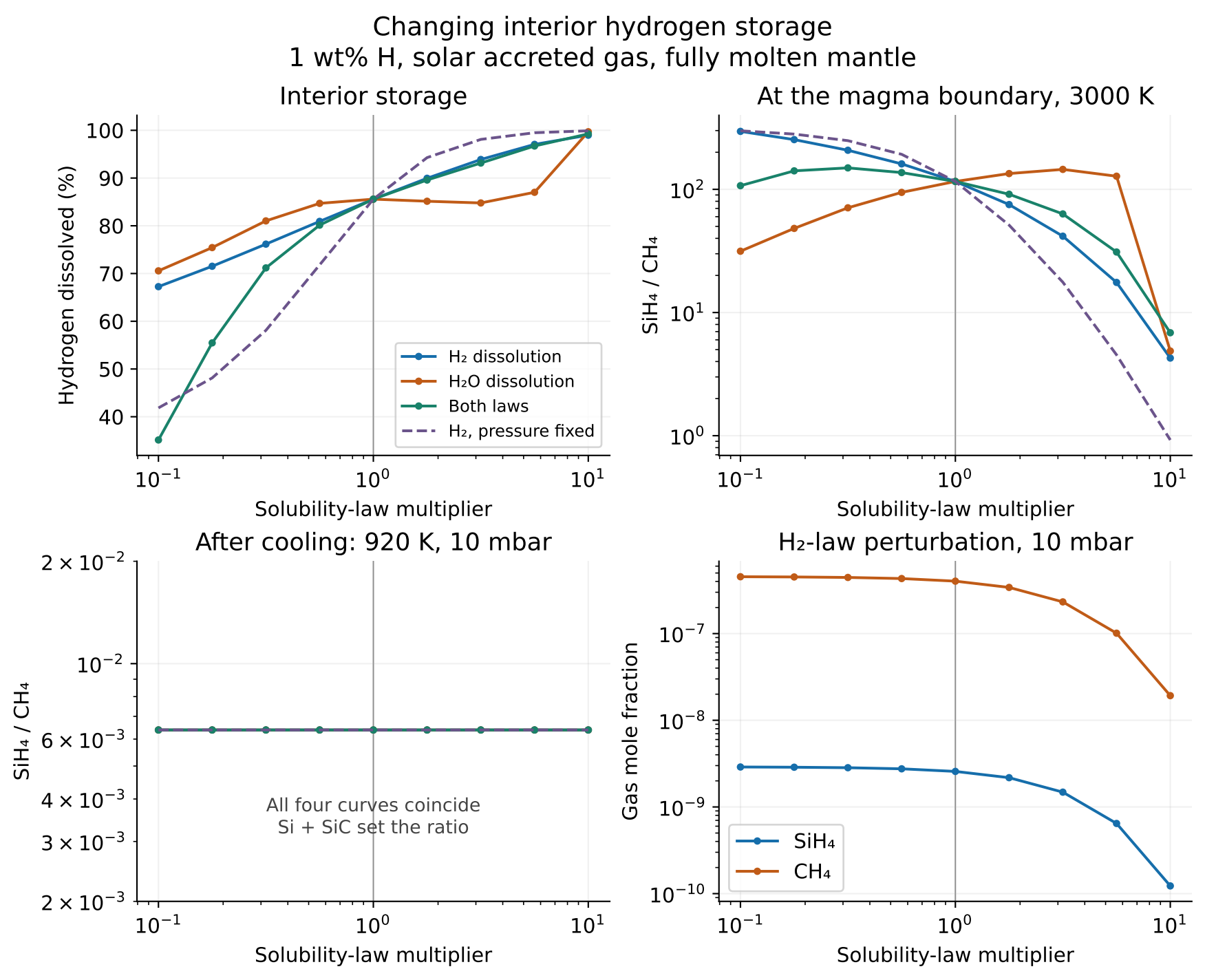
Changing interior hydrogen storage
1 wt% H, solar accreted gas, fully molten mantle
Interior storage
Hydrogen dissolved (%)
Solubility-law multiplier
H₂ dissolution
H₂O dissolution
Both laws
H₂, pressure fixed
At the magma boundary, 3000 K
SiH₄ / CH₄
Solubility-law multiplier
After cooling: 920 K, 10 mbar
SiH₄ / CH₄
All four curves coincide
Si + SiC set the ratio
Solubility-law multiplier
H₂-law perturbation, 10 mbar
Gas mole fraction
SiH₄
CH₄
Solubility-law multiplier